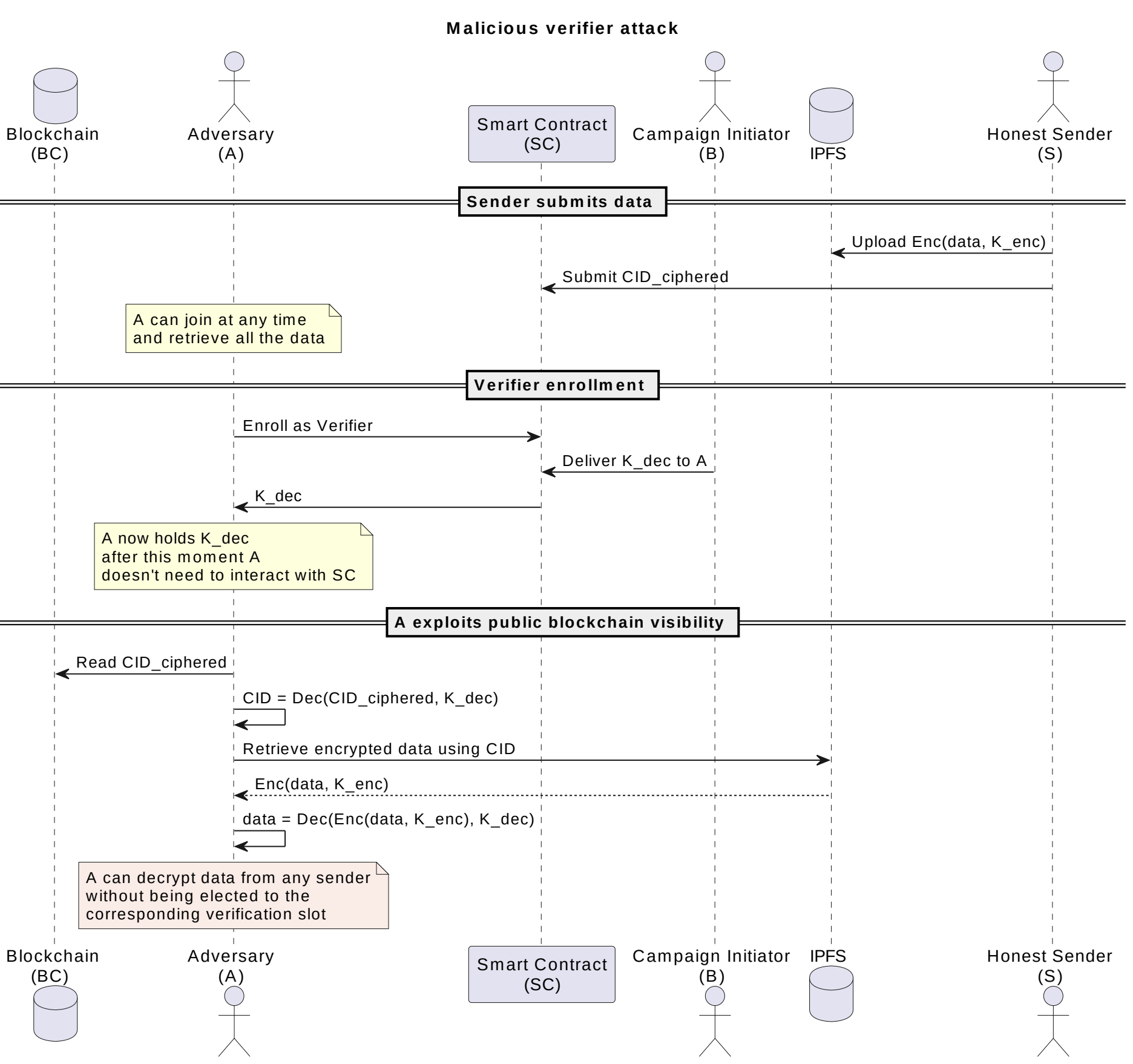
Malicious verifier attack
Blockchain (BC)
Adversary (A)
Smart Contract (SC)
Campaign Initiator (B)
IPFS
Honest Sender (S)
Sender submits data
Upload Enc(data, K_enc)
Submit CID_ciphered
A can join at any time and retrieve all the data
Verifier enrollment
Enroll as Verifier
Deliver K_dec to A
K_dec
A now holds K_dec after this moment A doesn't need to interact with SC
A exploits public blockchain visibility
Read CID_ciphered
CID = Dec(CID_ciphered, K_dec)
Retrieve encrypted data using CID
Enc(data, K_enc)
data = Dec(Enc(data, K_enc), K_dec)
A can decrypt data from any sender without being elected to the corresponding verification slot
Blockchain (BC)
Adversary (A)
Smart Contract (SC)
Campaign Initiator (B)
IPFS
Honest Sender (S)